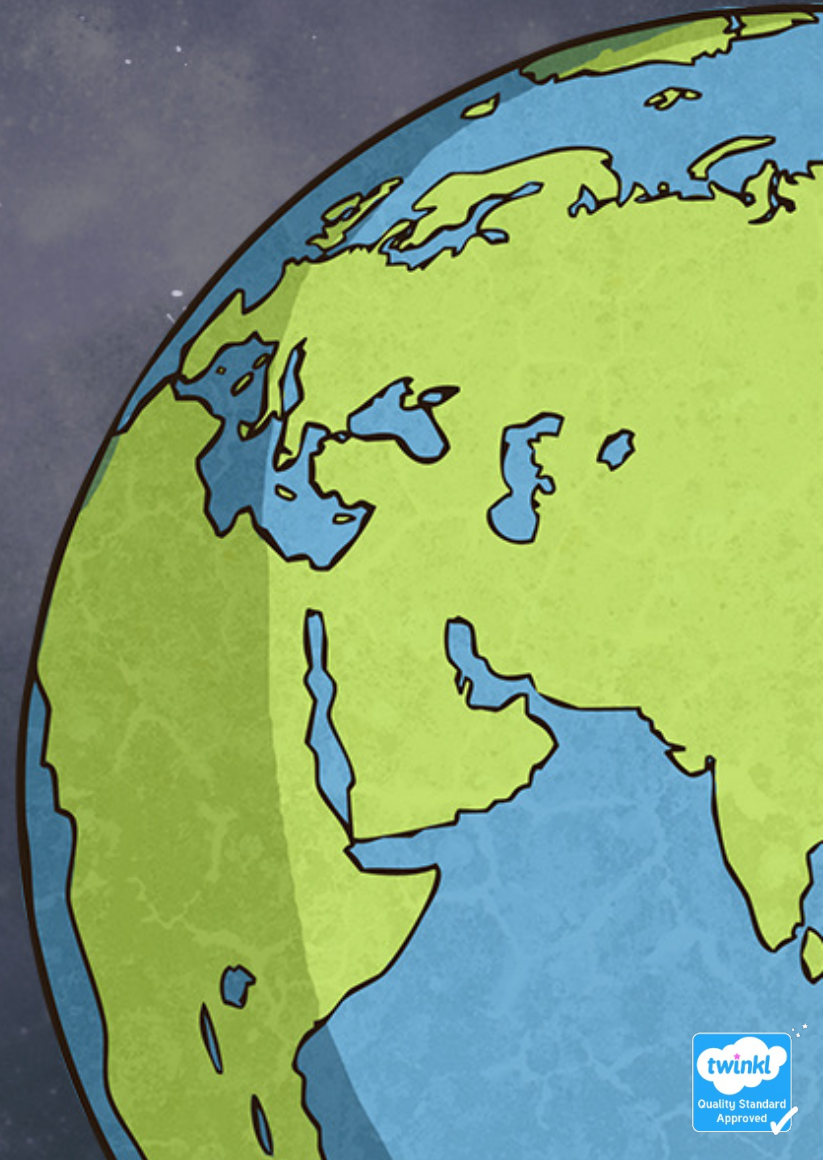


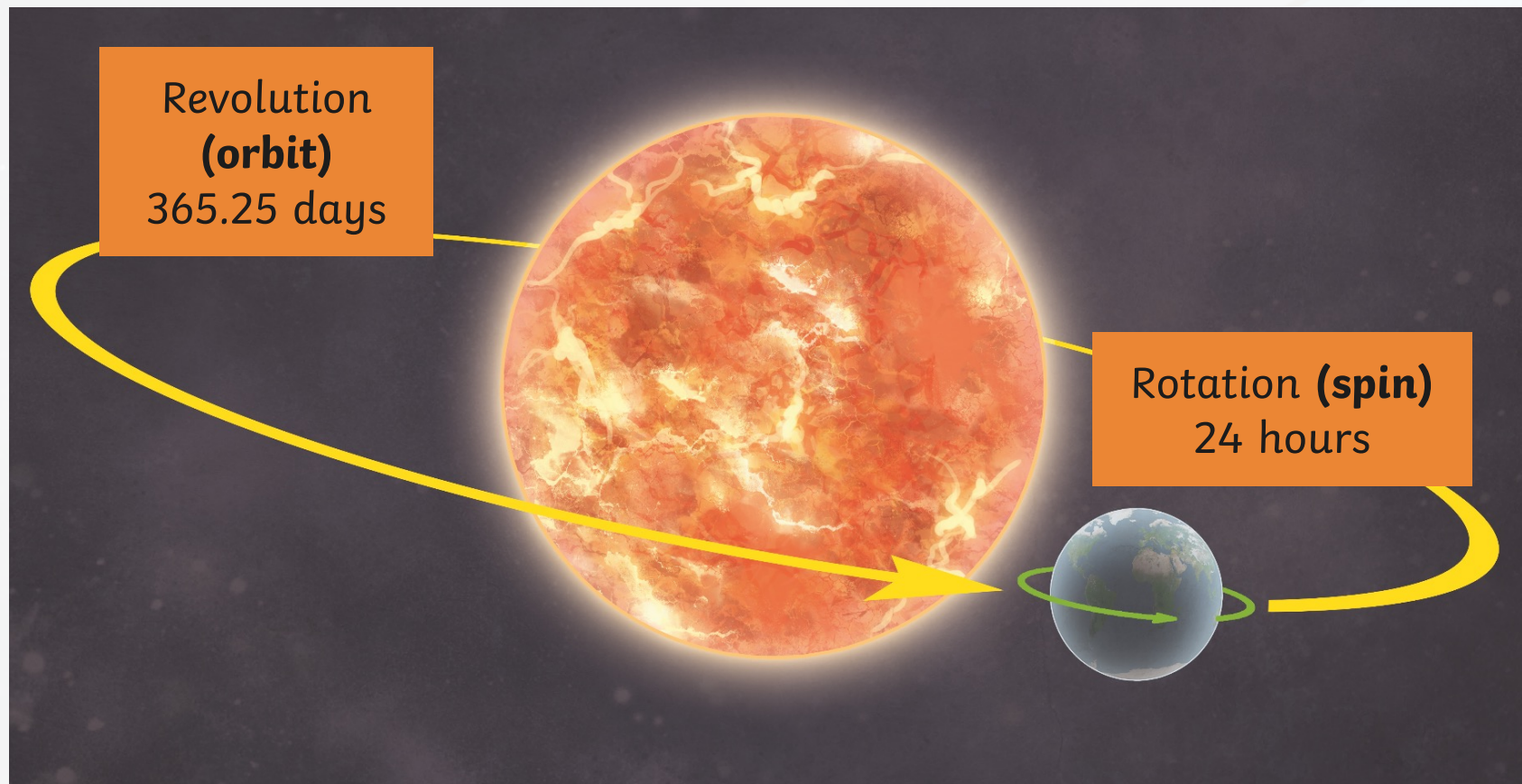
# Earth's Movement in Space

Rotation vs Revolution



# Earth's Movement

The Earth moves in two different ways in space.  
The Earth **rotates** and **revolves**.



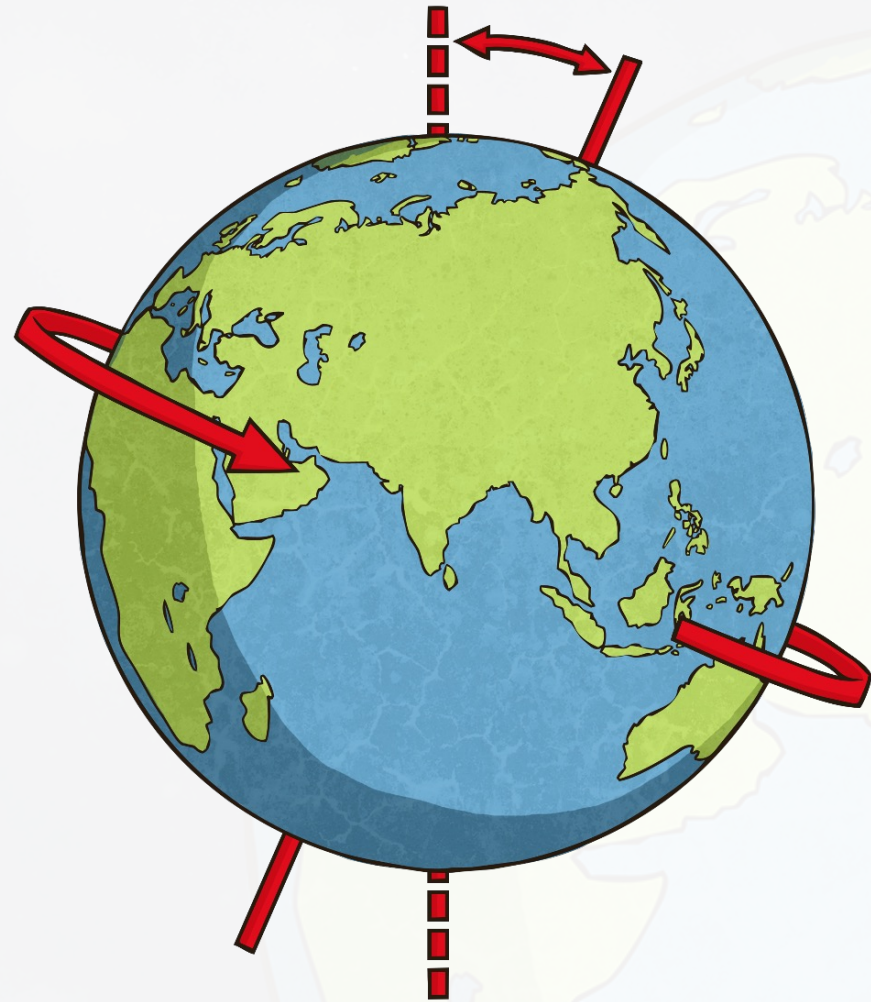


# Earth's Rotation

There is an imaginary line through the center of the Earth called the **axis**.

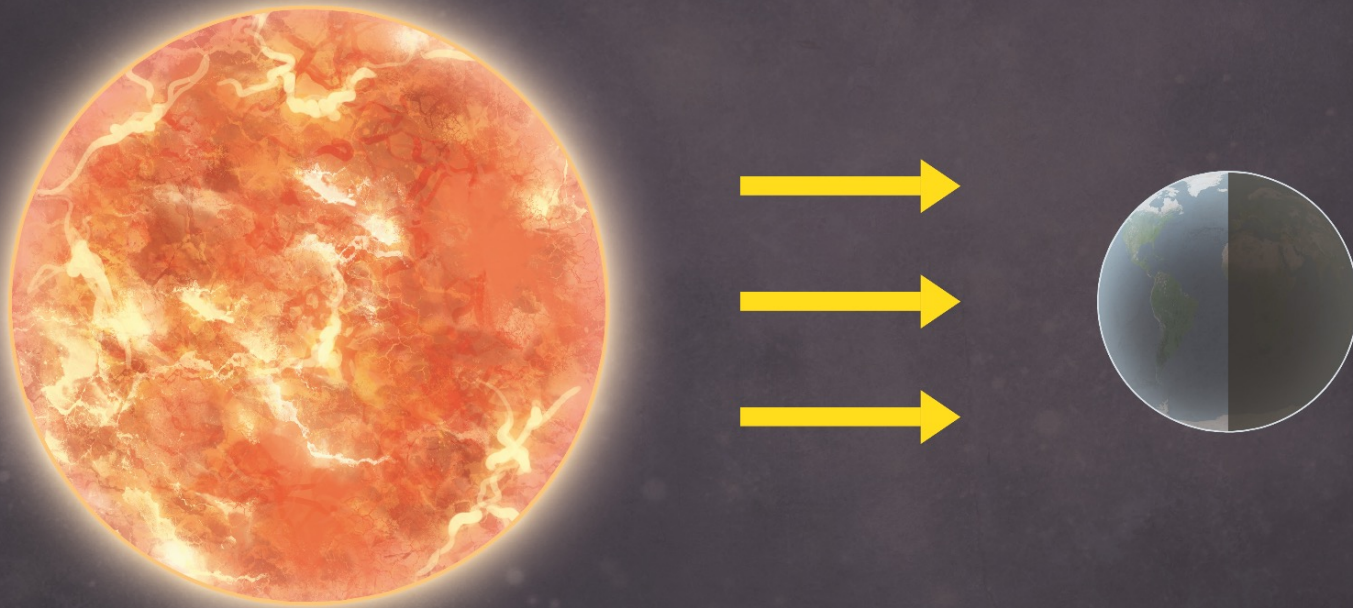
The axis extends from north to south.

The axis is at a tilt of 23.5 degrees.



# Earth's Rotation

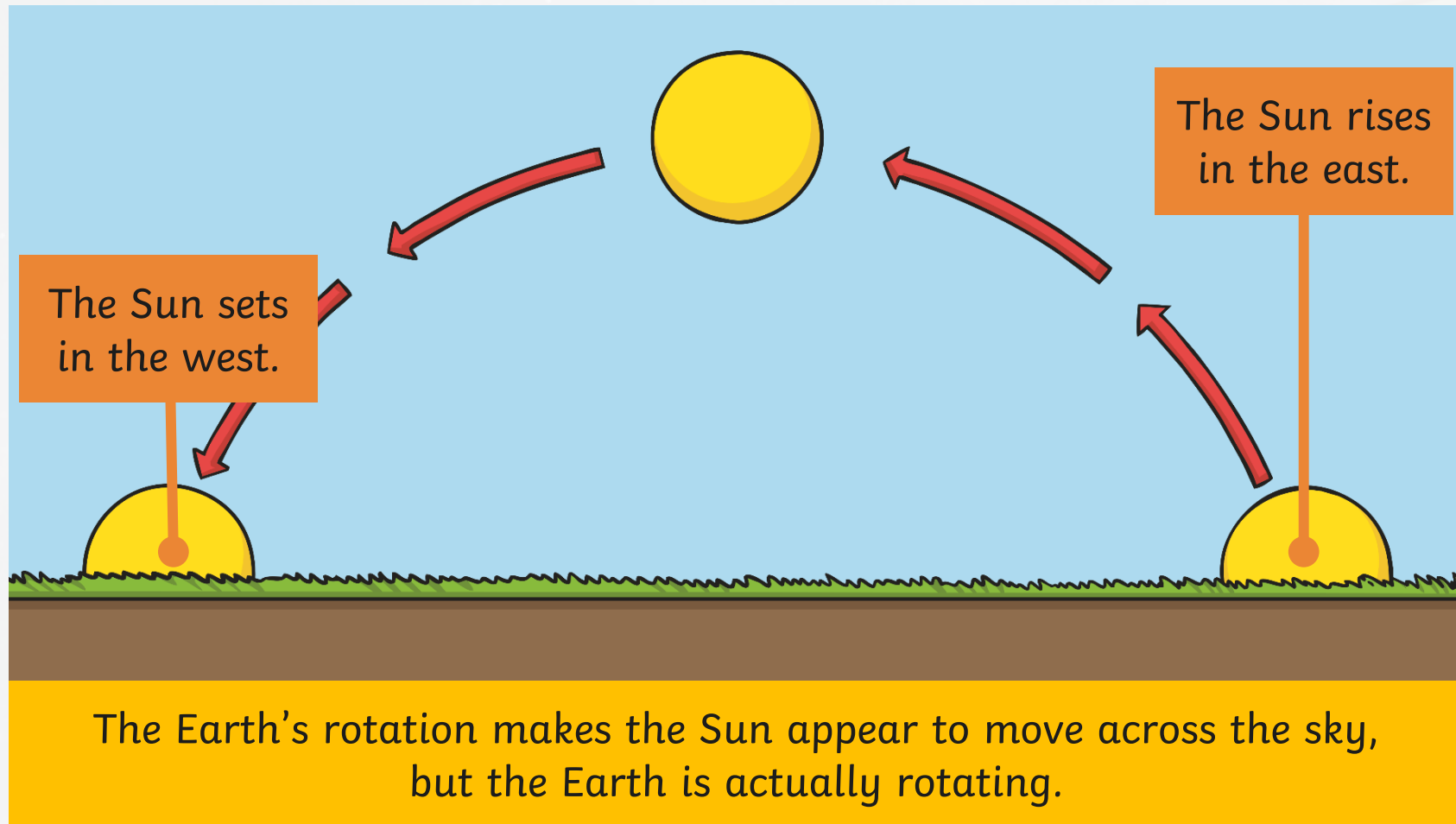
It takes the Earth 24 hours, or 1 day, to complete one rotation on its axis.



When the side of the Earth that is facing the Sun is experiencing **daytime**, the side of the Earth that is facing away from the Sun is experiencing **night-time**.

# The Sun

The Sun rotates on its axis once in about 27 days, but since it is not solid, the equator (25 days) rotates faster than the poles (36 days)

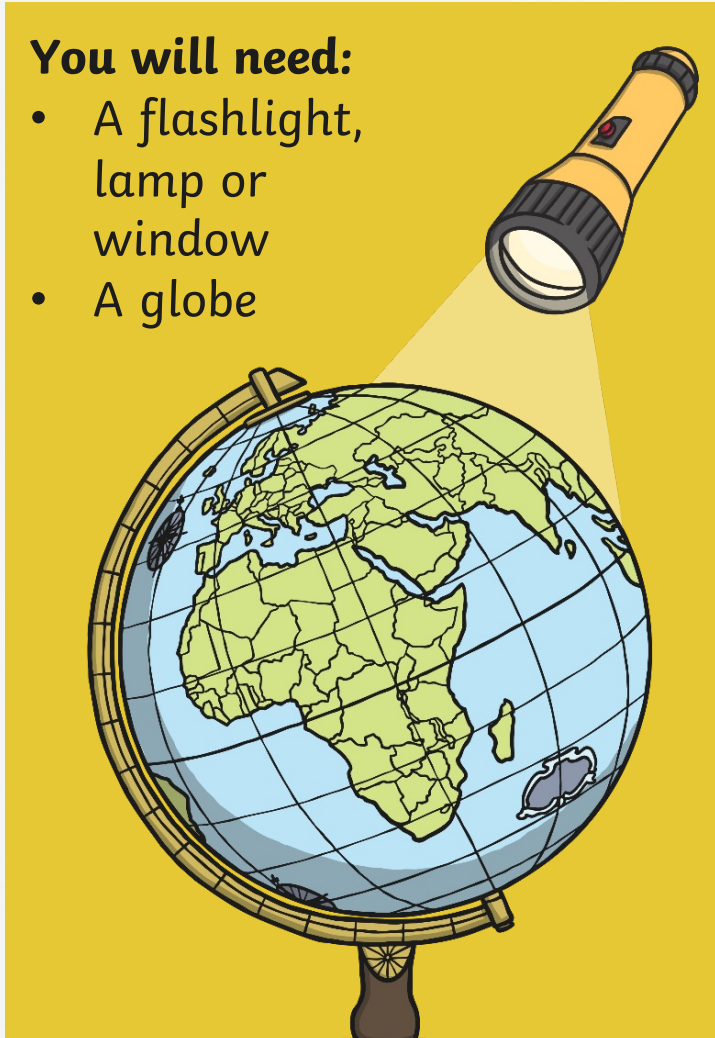




# Give It a Try

## You will need:

- A flashlight, lamp or window
- A globe



1. Hold the globe next to the light source.

2. Find where you live on the globe.

3. Turn off the lights.

4. Point to where the axis is on the Earth (remember the Earth is tilted on its axis).

5. Rotate the Earth counterclockwise to model day and night.

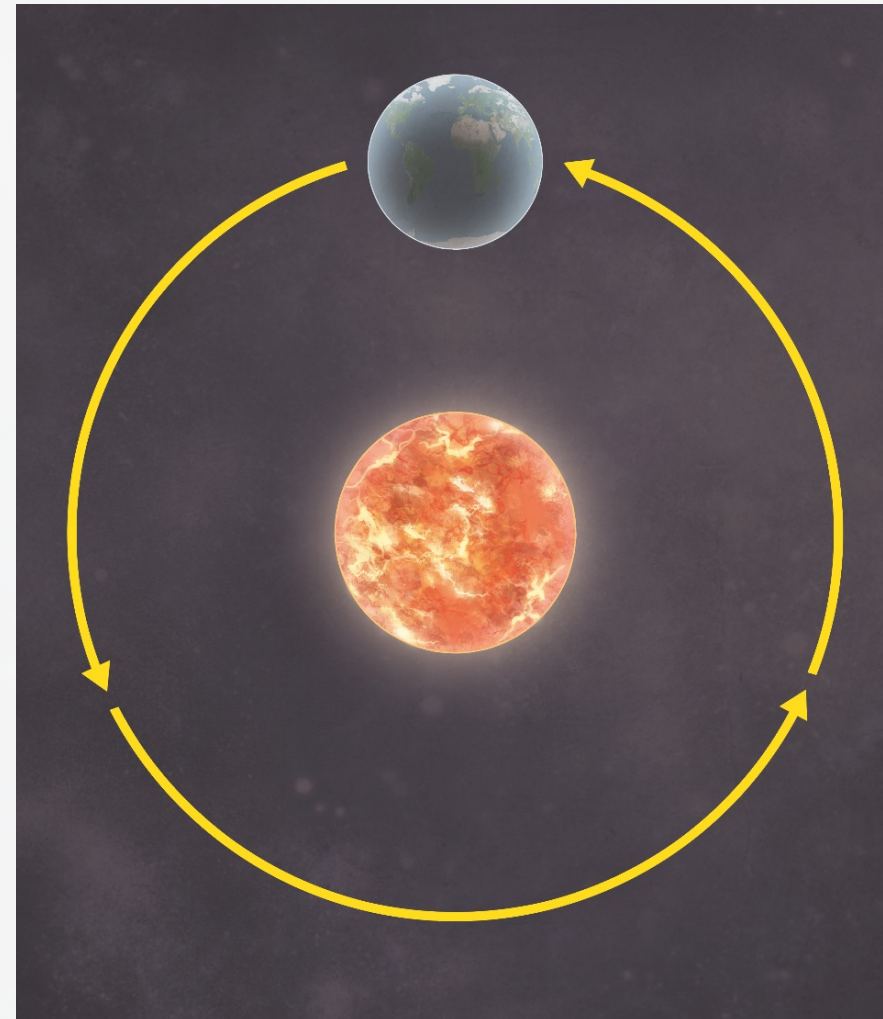
# Earth's Revolution

The Earth revolves around the Sun.

This takes approximately 365 days, or 1 year.

The path the Earth takes around the sun is called Earth's 'orbit'.

The Earth's orbit is in an elliptical shape, so it is not perfectly round.

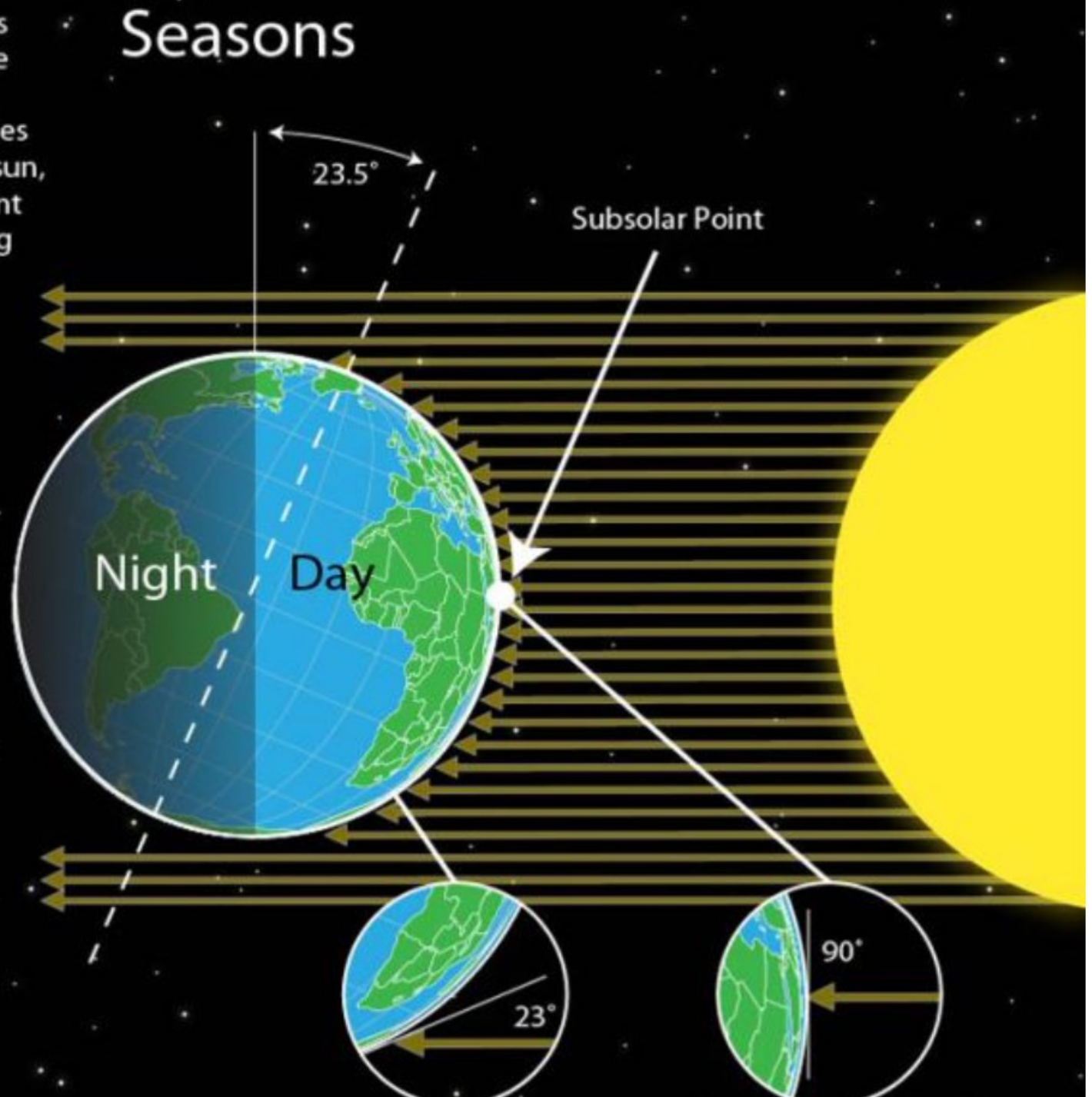




One of the most important factors that determines the seasons is the maximum intensity of the sun's rays over a given area. This changes as the Earth revolves around the sun, tilted on its axis. The subsolar point moves north and south, indicating summer in the Northern or Southern Hemisphere.

This illustration shows summer in the Northern Hemisphere. The Northern Hemisphere is tilted toward the sun, and the subsolar point is north of the Equator, where the sun's rays hit the Earth at a right angle. The angle of sunlight decreases as you move away from the subsolar point. The angle of sunlight reaching the South Pole, for instance, is almost 0. There, it is winter. As the Earth continues to revolve and the Southern Hemisphere tilts toward the sun, the Southern Hemisphere will experience summer.

(not to scale)

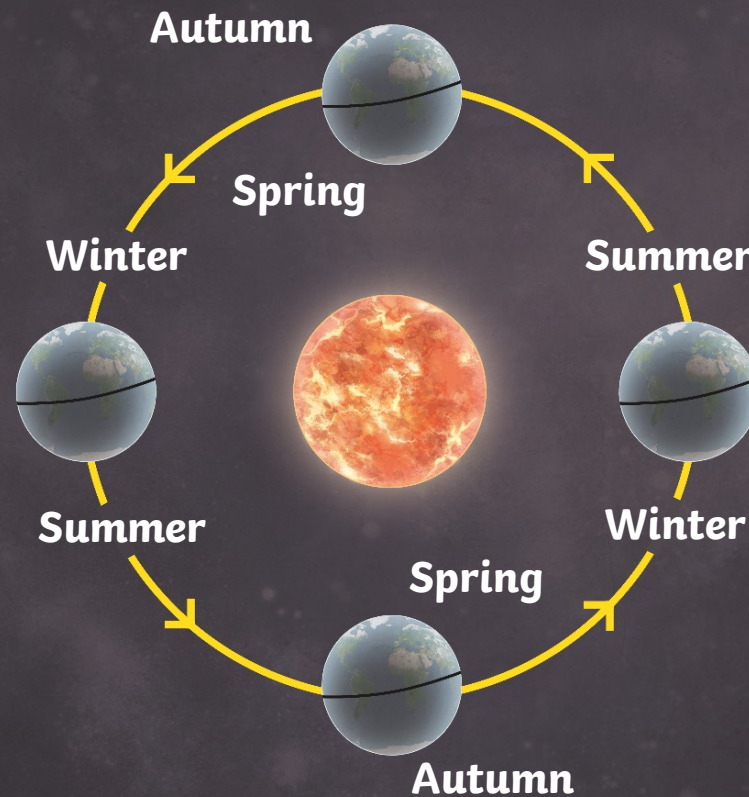




# Seasons in the Northern Hemisphere

The seasons are created through the combination of the Earth's **revolution** and the **tilt of the axis**.

When the northern hemisphere of the Earth is tilted away from the Sun, people in these regions are experiencing **winter**.

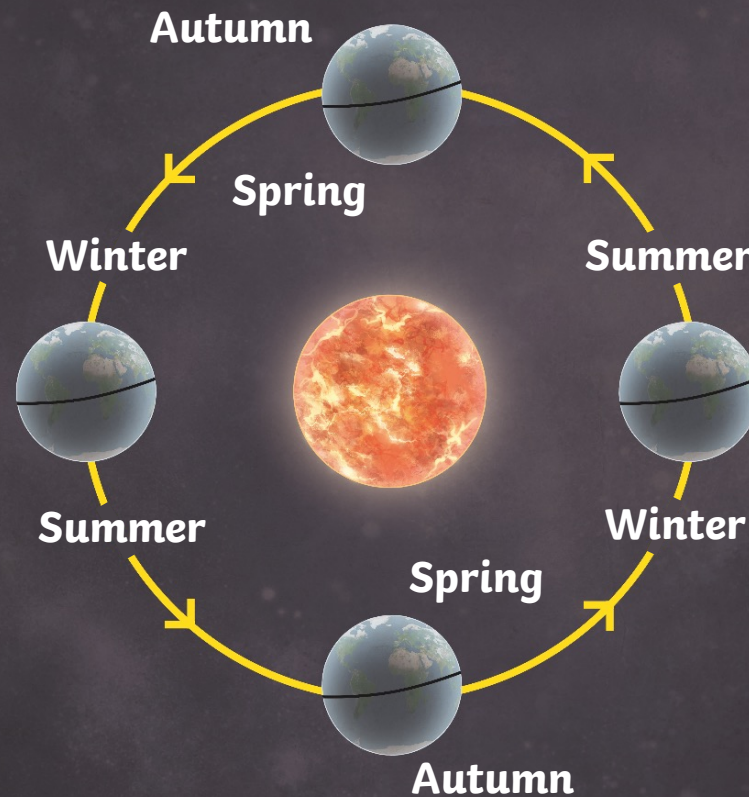


When the northern hemisphere of the Earth is tilted toward the Sun, people in these regions are experiencing **summer**.

# Seasons in the Southern Hemisphere

The seasons are created through the combination of the Earth's **revolution** and the **tilt of the axis**.

When the southern hemisphere of the Earth is tilted away from the Sun, people in these regions are experiencing **winter**.



When the southern hemisphere of the Earth is tilted toward the Sun, people in these regions are experiencing **summer**.



# Can You Revolve?



## You will need:

- A partner
- A little space to move

**1.** Have one friend represent the Earth, and the other friend represent the Sun.

**2.** The 'Sun' will stand still, while the 'Earth' revolves around the sun.

**3.** Switch roles with your partner and try it again.

# Do You Know the Difference?

Explain to your partner how **rotation** and **revolution** are different.  
Try to use as many of these key words as possible:

axis

spin

orbit

seasons

tilt

year

day

night



